

The Structure of Local Motion: Unpacking the State Transition Matrix

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Abstract

The local motion about a trajectory in a Hamiltonian Dynamical system, in particular in an astrodynamics system, is controlled by the state transition matrix. By analyzing this solution one can obtain insights into the local stability and instability of motion, can identify constraints on the volumetric propagation of solutions, and other key insights. In this talk we will introduce and discuss some of the fundamental structures that can be found in the state transition matrix and introduce a novel approach to analyzing general trajectories. Examples from the 2-body problem are given.

1 Introduction

A key component for analyzing motion in astrodynamics problem is the state transition matrix (STM), which describes how the local phase space maps in the vicinity of a given trajectory. There are a plethora of applications for the STM, including stability and instability analysis, computation of periodic and quasi-periodic motions, use in mapping uncertainty distributions, and determining sensitivity of a trajectory to state or parameter errors. In this note we analyze the fundamental structure associated with the STM in terms of the Hamiltonian system parameters from which it is derived. In particular, we introduce a recently derived ortho-symplectic transformation

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that can be applied to separate the basic flow components associated with an STM, enabling the definition of a local Poincaré-like map that can be defined between any two points on a trajectory, and which devolves to the classical Poincaré map and associated monodromy matrix when evaluated over a periodic motion.

2 Hamiltonian Dynamical System Basics

Recall that a Hamiltonian Dynamical system is described as

$$\dot{\mathbf{x}} = JH_{\mathbf{x}}(\mathbf{x})$$

where $H(\mathbf{x})$ is the Hamiltonian function, $\mathbf{x} = [\mathbf{q}, \mathbf{p}] \in \mathbb{R}^{2n \times 2n}$ and

$$J = \begin{bmatrix} \mathbf{0}_{n \times n} & \mathbf{I}_{n \times n} \\ -\mathbf{I}_{n \times n} & \mathbf{0}_{n \times n} \end{bmatrix}$$

Given a general state $\mathbf{x}_o$ and a time t_o , the resulting flow of the dynamical system driven by the Hamiltonian Dynamical System (HDS) can be described as $\mathbf{x}(t) = \phi(t; \mathbf{x}_o, t_o)$ and satisfies the HDS: $\partial\phi/\partial t = J \partial H/\partial \mathbf{x}|_{\phi}$.

The STM is defined as the partial of the solution flow with respect to the initial condition $\mathbf{x}_o$, $\Phi(t, t_o, \mathbf{x}_o) = \frac{\partial\phi}{\partial \mathbf{x}_o}$. The STM also satisfies the linearized differential equation

$$\begin{aligned} \dot{\Phi}(t, t_o) &= JH_{\mathbf{xx}}|_{\phi(t)}\Phi(t, t_o) \\ \Phi(t_o, t_o) &= I \end{aligned}$$

We note that a Hamiltonian system can be alternatively stated such that the state is instead ordered in coordinate-momentum pairs, or $\mathbf{z} = [q_1, p_1, \dots, q_n, p_n]$. To formally transform between the different Hamiltonian statements just requires a permutation matrix $P_{x \rightarrow z}$ defined to exchange rows i to $2i - 1$ for $i = 1, 2, \dots, n$, and rows $i + n$ to $2i$ for rows $n + 1, n + 2, \dots, 2n$. The resulting matrix is orthogonal, thus $P_{x \rightarrow z}^T = P_{z \rightarrow x}$, but is not symplectic. Thus, the symplectic matrix is redefined, applying the transformation, to $\mathcal{J} = P_{x \rightarrow z} J P_{z \rightarrow x}$, or

$$\mathcal{J} = \begin{bmatrix} J_2 & 0_{2 \times 2} & \cdots & 0_{2 \times 2} \\ 0_{2 \times 2} & J_2 & \cdots & 0_{2 \times 2} \\ 0_{2 \times 2} & \cdots & 0_{2 \times 2} & J_2 \end{bmatrix}$$

where

$$J_2 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

We will transform between these two formulations as needed in the following.

3 The Ortho-Symplectic Transformation

In [1] a procedure is introduced that derives an ortho-symplectic transformation that is based on the local dynamical flow evaluated at a state, $JH_{\mathbf{x}}$, and the Hamiltonian gradient $H_{\mathbf{x}}$. The procedure starts by defining a basis direction $\mathbf{u}_1 = J\hat{H}_{\mathbf{x}}$ as the first coordinate direction and $\mathbf{v}_1 = -J\mathbf{u}_1 = \hat{H}_{\mathbf{x}}$ as the first momentum direction, where the $\hat{\cdot}$ notation signifies a unit vector, specifically $\hat{H}_{\mathbf{x}} = H_{\mathbf{x}}/|H_{\mathbf{x}}|$. The remaining coordinate bases $\mathbf{u}_j$ are constructed to be orthogonal to the previous coordinates $\mathbf{u}_k$, $k < j$ using a Gram-Schmitt type of approach. Then the corresponding momentum bases are defined as $\mathbf{v}_j = -J\mathbf{u}_j$. Grouping the terms together defines the ortho-symplectic transformation that is defined at any state:

$$R(\mathbf{x}) = [\mathbf{u}_1 \mathbf{u}_2 \cdots \mathbf{u}_n \mathbf{v}_1 \mathbf{v}_2 \cdots \mathbf{v}_n]$$

In [1] it is shown that this transformation is both symplectic and orthogonal, meaning that $R^T = R^{-1}$ and $RJ = JR$. This transformation can be used to show that phase space in the $\mathbf{u}_1$ and $\mathbf{v}_1$ coordinates, correlated with the fundamental solution flow and gradient of the Hamiltonian at a given state, are uncoupled from the other directions and locally conserve phase volume, implying that the remaining bases also conserve volume separately. In this way the transformation separates the local phase volume into disjoint components. In this note we explore the full transformation and develop a concept for using it to simplify and expose the local structure of motion.

The transformation can be defined at any given state, $\mathbf{x}$, and the linear mapping about that state from an initial time t_o , state $\mathbf{x}_o$ and deviation $\delta\mathbf{x}_o$ to a final time t_f , state $\mathbf{x}_f$ and deviation $\delta\mathbf{x}_f$, where $\delta\mathbf{x}_f = \Phi(t_f, t_o)\delta\mathbf{x}_o$. Denote the transformations at the endpoint times as $R(\mathbf{x}_o) = R_o$ and $R(\mathbf{x}_f) = R_f$. Then we can define the transformed state deviations as $\delta\mathbf{X} = R^T\delta\mathbf{x}$. We note that the coordinate $\delta X(1)$ is associated with the solution flow and $\delta X(n+1)$ is associated with the local Hamiltonian gradient. Applying the transformation to the linear mapping changes $\delta\mathbf{x}_f = \Phi\delta\mathbf{x}_o$ into $\delta\mathbf{X}_f = \Psi\delta\mathbf{X}_o$, where $\Psi = R_f^T\Phi R_o$. We note that Ψ is symplectic as the original Φ and the

R transformations are all symplectic. The mapping Ψ goes between the basis evaluated at the initial state to the basis evaluated at the final state.

For a general Hamiltonian system we can develop a fully general ortho-symplectic transformation for any given state. First, we define the same initial coordinate and momentum directions:

$$\mathbf{u}_1 = J\hat{\mathbf{H}}_{\mathbf{x}} \quad (1)$$

$$\mathbf{v}_1 = \hat{\mathbf{H}}_{\mathbf{x}} \quad (2)$$

Here we can note that the gradient $H_{\mathbf{x}} = [H_{\mathbf{q}}^T, H_{\mathbf{p}}^T]^T$ and is split into the two components $H_{\mathbf{q}}$ and $H_{\mathbf{p}}$. Thus, we can rewrite the initial unit vectors as:

$$\mathbf{u}_1 = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} H_{\mathbf{p}} \\ -H_{\mathbf{q}} \end{bmatrix} \quad (3)$$

$$\mathbf{v}_1 = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} H_{\mathbf{q}} \\ H_{\mathbf{p}} \end{bmatrix} \quad (4)$$

It is also clear that these vectors have unit magnitude and are orthogonal.

For a 2-DOF system each of these sub-components is a 2-vector. We note that we can construct additional possible unit vectors of our transformation by inspection, given these sub-vectors. If we assume that the two components of each of these vectors can be identified with a generic frame denoted as $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$, where $\hat{\mathbf{x}} \cdot \hat{\mathbf{y}} = 0$. Then we can define the cross product of these vectors with the out-of-plane unit vector defined as $\hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}$. The operations $\hat{\mathbf{z}} \times H_{\mathbf{q}}$ and $\hat{\mathbf{z}} \times H_{\mathbf{p}}$ are then well defined and produce vectors that lie in the original $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ plane. We can then propose the following additional set of basis vectors for our ortho-symplectic transformation:

$$\mathbf{u}_2 = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} \hat{\mathbf{z}} \times H_{\mathbf{p}} \\ \hat{\mathbf{z}} \times H_{\mathbf{q}} \end{bmatrix} \quad (5)$$

$$\mathbf{v}_2 = -J\mathbf{u}_2 = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} -\hat{\mathbf{z}} \times H_{\mathbf{q}} \\ \hat{\mathbf{z}} \times H_{\mathbf{p}} \end{bmatrix} \quad (6)$$

where again these are all of unit magnitude and orthogonal to each other.

It is simple to note that $\mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{v}_1 \cdot \mathbf{v}_2 = 0$, due to the properties of the cross product. Further, by construction we know that $\mathbf{u}_1 \cdot \mathbf{v}_1 = \mathbf{u}_2 \cdot \mathbf{v}_2 = 0$. The other orthogonality relations, $\mathbf{u}_1 \cdot \mathbf{v}_2$ and $\mathbf{u}_2 \cdot \mathbf{v}_1$, need to be worked out in more detail. First note that

$$\mathbf{u}_2 \cdot \mathbf{v}_1 = -\mathbf{u}_1 \cdot \mathbf{v}_2 = (\hat{\mathbf{z}} \times H_{\mathbf{q}}) \cdot H_{\mathbf{p}} + H_{\mathbf{q}} \cdot (\hat{\mathbf{z}} \times H_{\mathbf{p}}) \quad (7)$$

Here we just need the triple scalar product identity: $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$. Applying this to the above we find that

$$\mathbf{u}_2 \cdot \mathbf{v}_1 = \hat{\mathbf{z}} \cdot (H_{\mathbf{q}} \times H_{\mathbf{p}} + H_{\mathbf{p}} \times H_{\mathbf{q}}) = 0 \quad (8)$$

Combining these forms the orthogonal matrix:

$$R = \frac{1}{|H_{\mathbf{x}}|} \begin{bmatrix} H_{\mathbf{p}} & \hat{\mathbf{z}} \times H_{\mathbf{p}} & H_{\mathbf{q}} & -\hat{\mathbf{z}} \times H_{\mathbf{q}} \\ -H_{\mathbf{q}} & \hat{\mathbf{z}} \times H_{\mathbf{q}} & H_{\mathbf{p}} & \hat{\mathbf{z}} \times H_{\mathbf{p}} \end{bmatrix} \quad (9)$$

Recall the test for symplecticity derived above, that an orthogonal matrix R is symplectic if $JR = RJ$. It is simple to show that this is satisfied for this transformation, defining our symplecto-orthogonal transformation. Finally, we note that the transformation is an explicit function of the Hamiltonian gradients, which in turn can be expressed as a function of the current state, or $R(\mathbf{x})$.

4 The Structure of Local Motion

In this new basis, the structure of the transformed state transition matrix has some unique and notable properties. Recall the definition

$$\Psi = R(\mathbf{x}_f)^T \Phi R(\mathbf{x}_o) \quad (10)$$

The STM as rewritten as Ψ will have some special structure due to the identities:

$$JH_{\mathbf{x}_f} = \Phi JH_{\mathbf{x}_o} \quad (11)$$

$$H_{\mathbf{x}_o}^T = H_{\mathbf{x}_f}^T \Phi \quad (12)$$

or in terms of the basis vectors,

$$|H_{\mathbf{x}_f}| \mathbf{u}_{1_f} = |H_{\mathbf{x}_o}| \Phi \mathbf{u}_{1_o} \quad (13)$$

$$|H_{\mathbf{x}_o}| \mathbf{v}_{1_o}^T = |H_{\mathbf{x}_f}| \mathbf{v}_{1_f}^T \Phi \quad (14)$$

The first identity means that the first column of Ψ has a special form:

$$R_f^T \Phi \mathbf{u}_{1_o} = \begin{bmatrix} \frac{|H_{\mathbf{x}_f}|}{|H_{\mathbf{x}_o}|} \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (15)$$

The second identity implies that the third row of Ψ has a special form:

$$\mathbf{v}_{1_f}^T \Phi R_o = \begin{bmatrix} 0 & 0 & \frac{|H_{\mathbf{x}_o}|}{|H_{\mathbf{x}_f}|} & 0 \end{bmatrix} \quad (16)$$

We note that both of these entries are guaranteed to be real and that they are inverses of each other. This was proven in [1] based on the symplectic structure of the system, and is not repeated here.

A final specific comment can also be made, that the term $\sigma_o = \mathbf{u}_{1_f}^T \Phi \mathbf{v}_{1_o} \neq 0$ in general. To prove this we note the following steps. First, $\mathbf{u}_{1_f} = \frac{1}{\lambda} \Phi \mathbf{u}_{1_o}$. Substituting this into the term gives us $\frac{1}{\lambda} \mathbf{u}_{1_o}^T \Phi^T \Phi \mathbf{v}_{1_o}$. Then, note that $\mathbf{u}_{1_o} = J \mathbf{v}_{1_o}$ and that $J \Phi^T = \Phi^{-1} J$. Thus we find that $\sigma_o = -\frac{1}{\lambda} \mathbf{v}_{1_o} \Phi^{-1} J \Phi \mathbf{v}_{1_o}$. However, this is generically non-zero unless $\Phi^{-1} = \Phi^T$, or that Φ is orthogonal, which is generally untrue. Thus, we find that $\sigma_o \neq 0$ in general.

As noted above, an equivalent form for a symplectic system is to group the coordinates and momenta together in pairs, thus making $\mathbf{z} = [q_1, p_1, q_2, p_2]$. The symplectic matrix is then comprised of small 2-D J matrices along the diagonal. This transformation can be applied to our systems by transforming with a permutation matrix that shifts entries 2 and 3. Doing this then gives a simplified and useful form for the $\tilde{\Psi} = P_{x \rightarrow z} \Psi P_{z \rightarrow x}$ matrix, $\delta \mathbf{z}_f = \tilde{\Psi} \delta \mathbf{z}_o$:

$$\tilde{\Psi} = \begin{bmatrix} \lambda & \sigma_o & \sigma_1 & \sigma_2 \\ 0 & 1/\lambda & 0 & 0 \\ 0 & \gamma_1 & a & b \\ 0 & \gamma_2 & c & d \end{bmatrix} \quad (17)$$

where $\lambda = \frac{|H_{\mathbf{x}_f}|}{|H_{\mathbf{x}_o}|}$, $\sigma_o = \mathbf{u}_{1_f}^T \Phi \mathbf{v}_{1_o}$, $\sigma_1 = \mathbf{u}_{1_f}^T \Phi \mathbf{u}_{2_o}$, $\sigma_2 = \mathbf{u}_{1_f}^T \Phi \mathbf{v}_{2_o}$, $\gamma_1 = \mathbf{u}_{2_f}^T \Phi \mathbf{v}_{1_o}$ and $\gamma_2 = \mathbf{v}_{2_f}^T \Phi \mathbf{v}_{1_o}$. Finally, we identify the lower right matrix as

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad (18)$$

We know that the matrix $\tilde{\Psi}$ is symplectic, by construction. Further, we can show that the sub-matrix M is symplectic. To establish this, we note that since the matrix $\tilde{\Psi}$ is symplectic its determinant must equal unity. Applying the general adjoint formula along the top row of the matrix, we find that

$$\det(\tilde{\Psi}) = \lambda \frac{1}{\lambda} \det(M) = \det(M) = 1 \quad (19)$$

Since the symplectic condition for a 2 by 2 matrix is just that it have unity determinant, we see that M is symplectic.

We can also place constraints on the σ_i and γ_i terms, due to the symplectic structure. Taking the above form of the resulting matrix Ψ , we can evaluate the symplectic condition to expose an additional constraint that must hold. Specifically, we can show that the σ_i and γ_i terms can be related to each other. The results are

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix} = -M^T J_2 \begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix}$$

and conversely

$$\begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} = M J_2 \begin{bmatrix} \sigma_1 \\ \sigma_2 \end{bmatrix}$$

It is interesting to note that we find, approximately, that $\sigma_1 \sim -\gamma_2$ and $\sigma_2 \sim -\gamma_1$, however there does not seem to be a strict equality.

5 The eigenstructure of Ψ and Φ

The above general form for the matrix $\tilde{\Psi}$ shows that the eigenstructure of $\tilde{\Psi}$ can be decoupled into the different coordinate-momentum pairs. Let us build up this result specifically. First we find that the eigenvalue condition becomes

$$(\mu - \lambda)(\mu - 1/\lambda) |\mu I_{2 \times 2} - M| = 0$$

For the eigenvalue $\mu = \lambda$ the eigenvector is $[1, 0, 0, 0]^T$. For the eigenvalue $\mu = 1/\lambda$ the eigenvector is $[0, 1, 0, 0]^T$. For the M matrix the eigenvalue condition reduces to just a function of the diagonal entries

$$\mu^2 - (a + d)\mu + 1 = 0$$

Thus provides the familiar result that the system is stable (has its eigenvalues on the unit circle) if $-2 < a + d < 2$, and is unstable with two reciprocal real roots if $|a + d| > 2$. In general we find

$$\mu_{\pm} = \frac{1}{2}(a + d) \left[1 \pm \sqrt{\frac{1}{4}(a + d)^2 - 1} \right]$$

The eigenvectors of these roots, given eigenvalues $\sigma_{\pm}$, are:

$$\begin{bmatrix} \frac{1}{\mu_{\pm}-\lambda} [\sigma_1 e_{\pm} + \sigma_2 f_{\pm}] \\ 0 \\ e_{\pm} \\ f_{\pm} \end{bmatrix}$$

where $e_{\pm}$ and $f_{\pm}$ satisfy the equivalent relationships $(\mu_{\pm} - a)e_{\pm} = bf_{\pm}$ or $(\mu_{\pm} - d)f_{\pm} = ce_{\pm}$.

We note that the eigenvalues and eigenvectors of Ψ will be different than those of the original STM Φ . This is because the transformation defined here is not a similarity transformation, as R_o and R_f are different, even if both are symplectic. However, if the nominal solution is periodic and evaluated over one period, then these transformation matrices will be equal and, in that case, Ψ and Φ will share eigenvectors and values. Additionally, we note that $\lambda = 1$ in that case, and that the eigenvalues of M will then determine the stability properties of the solution.

6 Solution Deviation

One common use of the state transition matrix is to use it to characterize the overall deviation of an initial condition in the vicinity of a nominal solution. This can be formally developed, but essentially it just devolves to the computation of the eigenvalues and eigenvectors of the matrix $\Phi^T \Phi$. Here, it is of interest to transform and consider the product of the Ψ matrix, as we can note that

$$\Psi^T \Psi = R_o^T \Phi^T \Phi R_o$$

Since this is now a similarity transformation, we note that the eigenvalues will be preserved, no matter what the final condition is. Performing the multiplication, the main takeaway is that the deviation is highly sensitive, and perhaps dominated by, the correlation terms σ_1 , σ_2 , γ_1 and γ_2 .

7 Local Poincaré Map

One use of this transformation is to define a Poincaré map between two arbitrary points along a trajectory. In a standard surface of section approach the monodromy matrix mapping fixes the Hamiltonian value and removes

variations along the trajectory. In the reduced map in the $\mathbf{z}$ state ordering we can enforce the energy conservation by setting $\delta p_{1_o} = 0$ and the initial starting surface orthogonal to the trajectory by setting $\delta q_{1_o} = 0$. Then the remaining coordinates δq_2 and δp_2 represent the local orbit variation with the energy and flow direction removed, thus representing a monodromy matrix.

With zero variations in energy the final variation in energy is also zero, or $\delta p_{1_f} = 0$. Even with the initial zero variations in energy and downtrack, however, the final variation in downtrack is not zero but will equal

$$\delta q_{1_f} = \sigma_1 \delta q_{2_o} + \sigma_2 \delta p_{2_o}$$

To force the final Poincaré map to lie along the single surface, we must allow the mapping time to vary as a function of the initial variations in the δq_2 and δp_2 . Note that the flow rate is precisely the state $q_{1_f} = |JH_{\mathbf{x}_f}|$, and thus we just need to adjust the time variation as:

$$\delta t_f = -\frac{1}{q_{1_f}} [\sigma_1 \delta q_{2_o} + \sigma_2 \delta p_{2_o}]$$

An interesting fact for this Poincaré Map definition is that over this short interval variation the other states are not expected to change at first order, since the q_1 direction contains the flow information and is orthogonal to the other directions. Thus, the map can be defined as

$$\delta \mathbf{z}_{2_f} = M \delta \mathbf{z}_{2_o}$$

which is independent of the time variation, which is only needed to be computed after the mapping.

Finally, if the nominal trajectory is evaluated relative to a periodic orbit over a full period, this reduced map represents the associated monodromy matrix.

Yet to be studied is whether this Poincaré Map formulation can be used to compute periodic orbit families.

References

- [1] O. Boodram and D.J. Scheeres. 2025. “Decoupled Phase Space Flexing in Autonomous Hamiltonian Systems,” IEEE Control Systems Letters 9: 643 - 648. DOI: 10.1109/LC-SYS.2025.3576069

- [2] F.Y. Hsiao and D.J. Scheeres. 2006. “Evolution of Eigenvalues during the Transient Period for Hamiltonian Systems,” *Physica D: Nonlinear Phenomena* 213: 66-75.